

Dataset link:

<https://www.kaggle.com/datasets/sabahezaraki/breast-ultrasound-images-dataset>

Overview of the Dataset:

Images of breast ultrasounds taken from women aged 25 to 75 were included in the baseline dataset. This information was arranged in 2018. Six hundred women make up the dataset. 780 photos with an average pixel size of 500*500 make up the dataset. However the dataset contains in total 1578 images including Ultrasound and Masked images In PNG format. The original photos are displayed beside the ground truth photos. The pictures fall into three categories: benign, malignant, and normal. The dataset contains some Ultrasound images with two affected places and that makes 2 masked images.

Data Cleaning:

Now, to clean the data first we merge those masked images that contain two affected places, so that they become one and use that image to train our model. The figure below shows, what was done:



Furthermore, as all the images are not of same size, I resized them to 256x256.

Dataset Metadata

Example of metadata of one image:

Image ID:

Dimension: 562x471

Width: 562 px

Height: 471 px

Horizontal Resolution: 96 DPI

Vertical Resolution: 96 DPI

Bit Depth: 24

File:

Item type: PNG File

Size: 326 KB

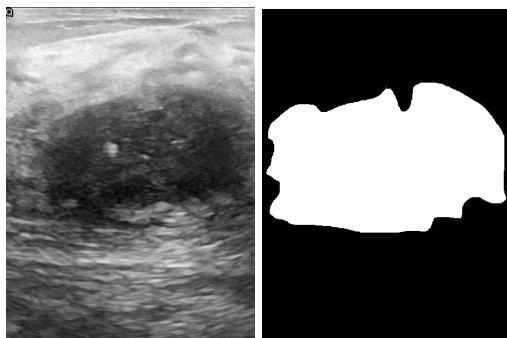
Attributes: A

Sample Images

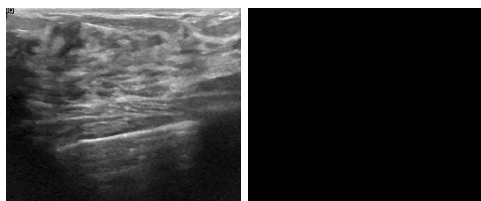
Below there are some sample images from each class:



Benign Class



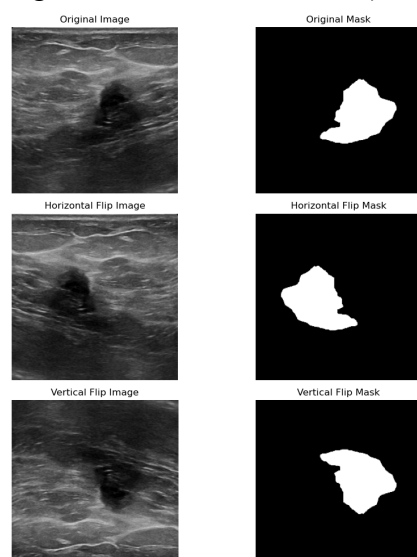
Malignant Class



Normal Class

Augmentation

As the number of images are very less, I augmented the data using techniques like Horizontal Flip and Vertical Flip. Doing so we increased the number of Ultrasound images and masked images to 1000 (We could augment to more but due to computational resources we augmented in small numbers). The image below shows an example of augmentation.



Tools & Libraries

Tensorflow, numpy, keras and matplotlib

Potential enrichment strategies include

- Collect more data from medical centers and approve their masked images.
- Use of GAN or Diffusion model to generate images and increase the size of the dataset.